

Interview Task: Processing GPS Data for Roundabout Detection

1. Task Background

Our software development requires processing large amounts of real-world environmental data. In this task, you need to develop a data processing system that identifies segments where an autonomous vehicle drives through a roundabout using GPS sensor data.

The GPS data is stored in a binary file, which is indexed using an SQLite database to facilitate efficient searching and retrieval.

Note: To simplify the task, we assume that the GPS points have approximately uniform density in the dataset. The timestamps do not reflect real-world movement but are generated based on a fixed average speed, as if the vehicle were traveling at a constant speed along the track. Therefore, timestamps, acceleration, and deceleration should not be considered in the solution.

2. Data Formats

2.1 GPS Data Binary File (gps_data.bin)

The binary file contains GPS data in the following structure:

Header (16 bytes)

Offset	Name	Size (bytes)	Description
0x00	version	4	File format version
0x04	count	4	Number of records
0x08	crc	8	Calculated using SHA-256 hash (64-bit CRC)

Records (40 bytes per record)

Offset	Name	Size (bytes)	Description
0x00	record_id	8	Unique identifier
0x08	synced_timestamp	8	Synchronized timestamp (nanosecond epoch)
0x10	device_timestamp	8	Sensor timestamp (microsecond epoch)

0x18	latitude	8	Latitude (degrees)
0x20	longitude	8	Longitude (degrees)
0x28	altitude	4	Altitude above sea level (meters)

2.2 SQLite Index File (*sensor_index.db*)

This file indexes the position of each packet in the binary data file. Schema: Packet (packet_id, sensor_id, synced_timestamp, offset, size) Sensor (sensor_id, name, filename)

Example: The {name} sensor's {packet_id} packet is located in {filename} starting at byte {offset} and spans {size} bytes.

3. Task

Implement a program that:

1. **Reads GPS data** from the binary file using the SQLite index file.
2. **Filters out invalid data:**
 - a. *Packet loss*: If packet_id is not continuous.
 - b. *Synchronization error*: If the difference between device_timestamp and synced_timestamp exceeds 15 microseconds.
 - c. *GPS error*: If the jump between GPS coordinates exceeds 1 degree or the values are invalid.
3. **Interpolates data** if 5 or more consecutive packets are invalid based on the above criteria.
4. **Detects roundabout traversal** based on GPS coordinates: Find an appropriate algorithm that can reliably detect when the vehicle is moving through a roundabout.

4. Output

The program should output the time intervals (along with corresponding GPS coordinates) when the vehicle was detected in a roundabout.

5. Expected Solution

The solution should be implemented in Python 3.11 and adhere to the following principles:

- **Efficient file handling** and binary reading.
- **Data cleaning and interpolation handling.**
- **Roundabout detection using GPS coordinates.**
- **Clean, readable, and well-documented code.**
- **Memory-efficient and optimized solution.**
- **Fault tolerance and robustness:** The program should handle anomalies and unexpected situations.
- **Scalability:** The solution should efficiently handle large datasets.